Lab 3

Visualizations and Performance Graphs

Field	Details
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Course	BCA(AI & DS)
Section	A
Lab Name	Lab 3 - Smart Resource Allocation and Scheduling System using Greedy Algorithms

1. Resource Allocation Chart (Fractional Knapsack)

Items: Gold (w=10, v=60), Silver (w=20, v=100), Bronze (w=30, v=120); capacity = 50. Gold and Silver are taken fully and 66.7% of Bronze, giving the maximum profit of 240.

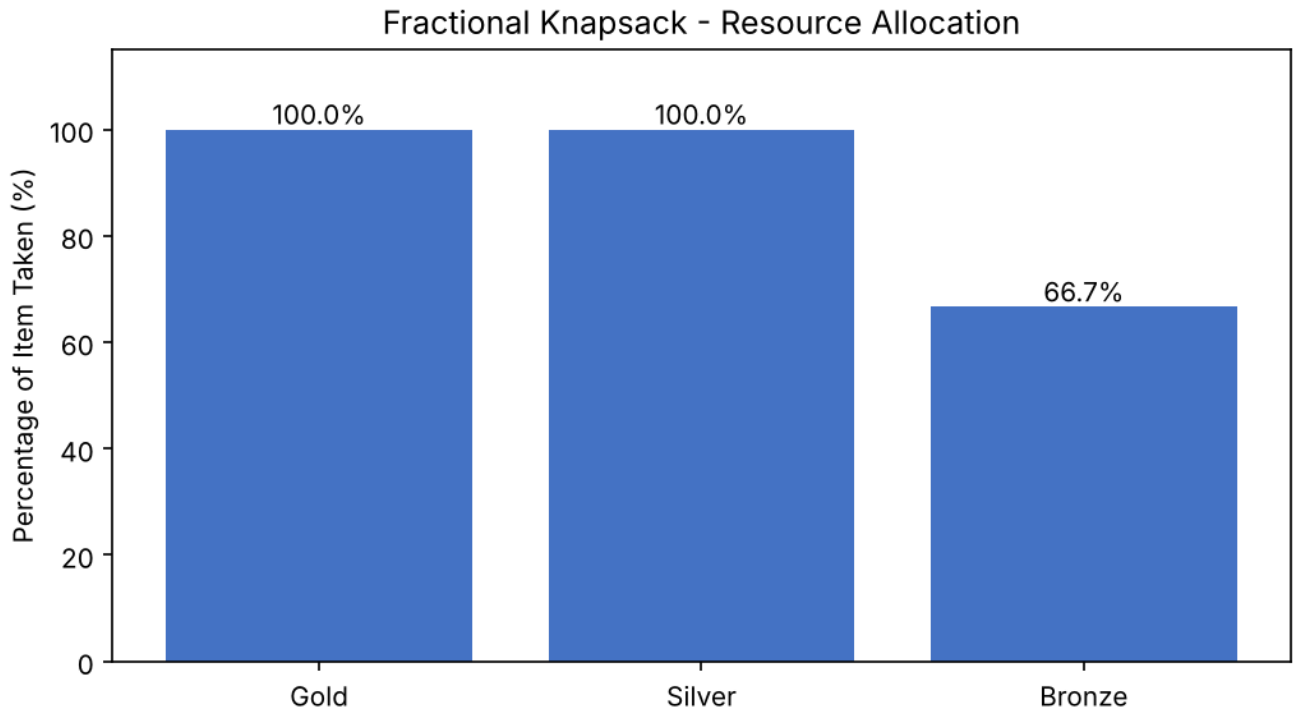


Figure 1: Fractional Knapsack - percentage of each item taken

2. Activity Timeline (Activity Selection)

Green bars are the selected non-overlapping activities (A1, A4, A8); grey bars were rejected because they overlap with an earlier selected activity.

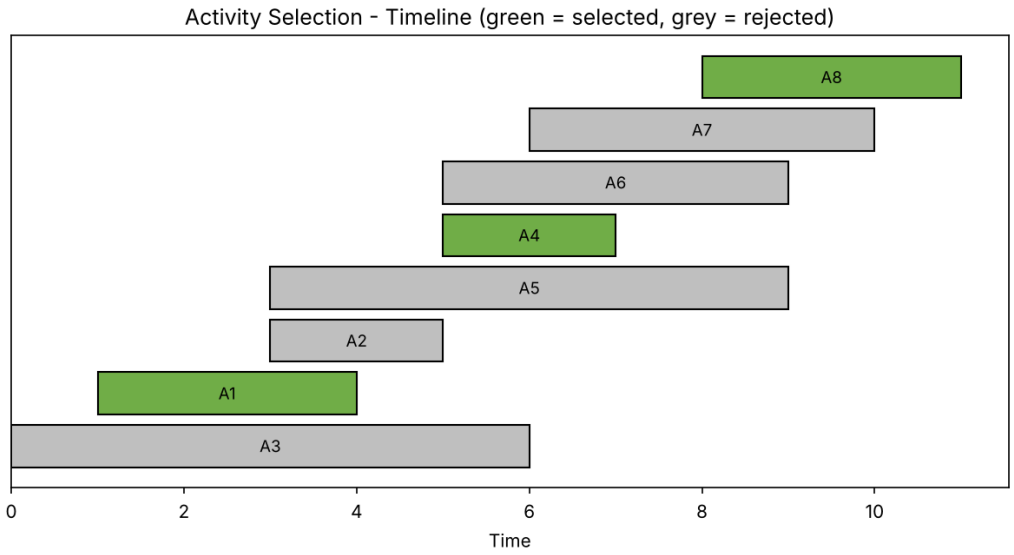


Figure 2: Activity Selection - scheduled timeline

3. Huffman Tree

Tree for the text 'abracadabra'. Left edge = 0, right edge = 1. Leaves show the character and its frequency; internal nodes show the combined frequency. Codes: a=0, d=100, c=101, r=110, b=111.

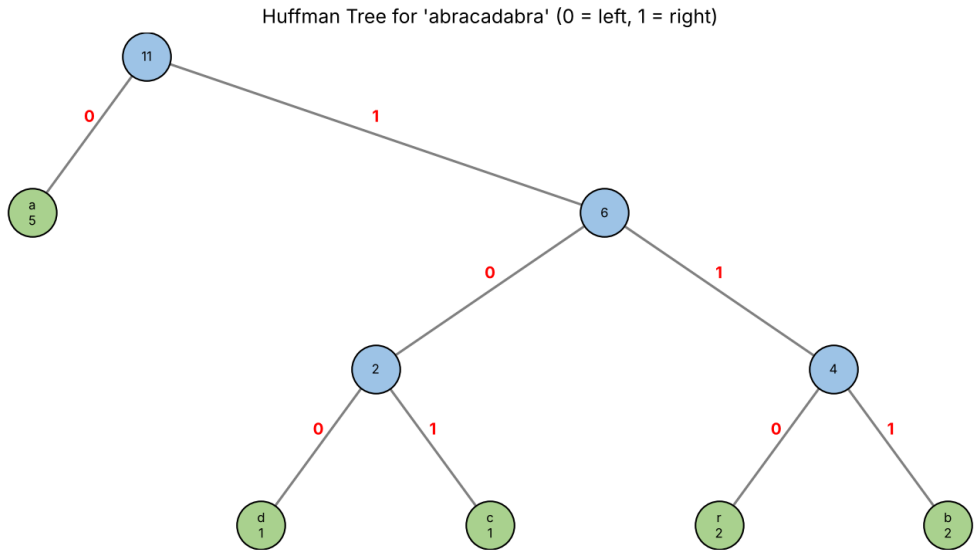


Figure 3: Huffman tree representation

4. Performance Graphs

Execution time versus input size for all three algorithms.

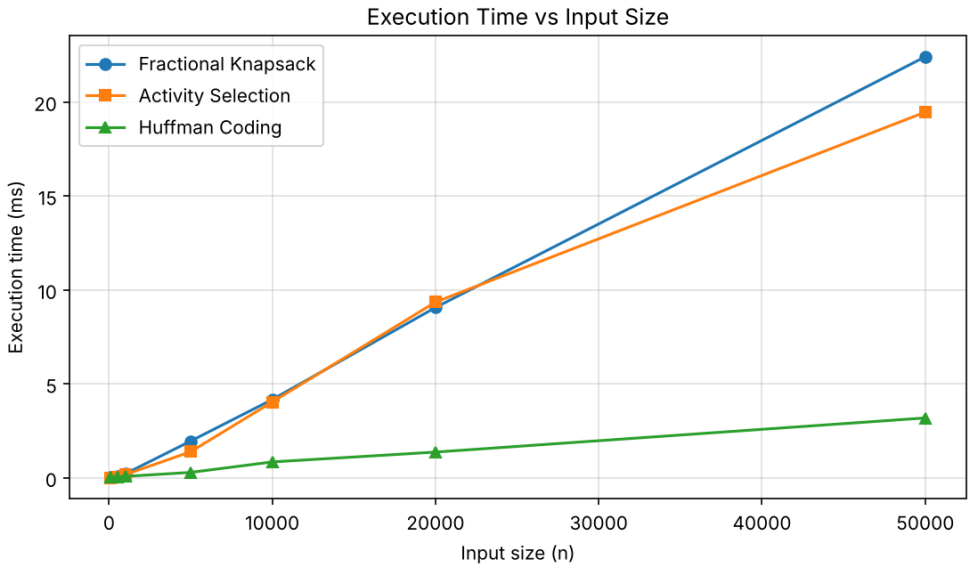


Figure 4: Execution time vs input size

Peak memory usage versus input size for all three algorithms.

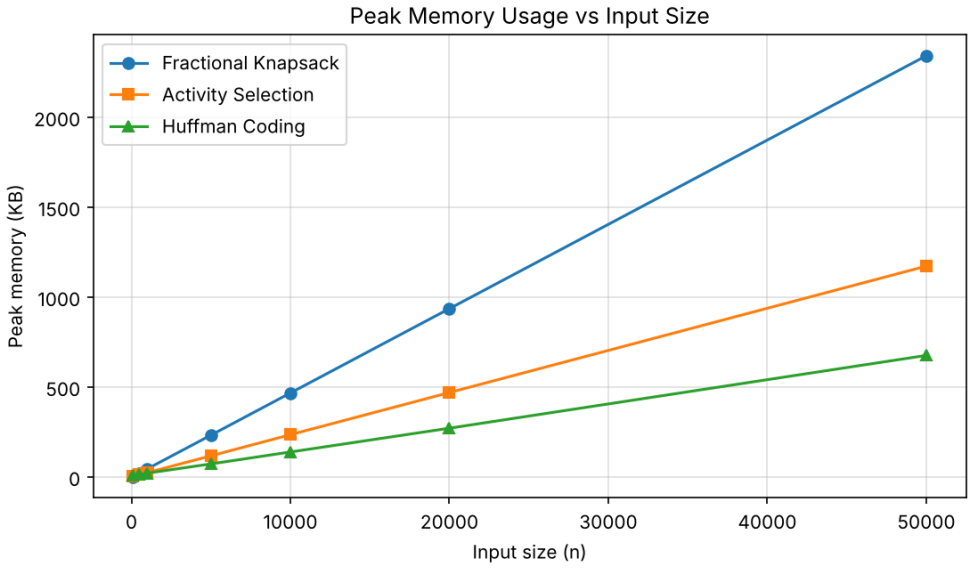


Figure 5: Peak memory usage vs input size

5. Key Observations

- The timeline clearly shows how choosing the earliest finishing activity leaves room for more activities.
- Time and memory curves for Knapsack and Activity Selection are close to linear on this scale, with a small n log n factor from sorting.
- Huffman Coding grows more slowly because its heap size depends on the number of distinct characters.